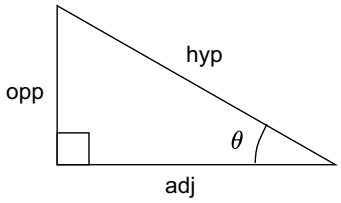
Trigonometry

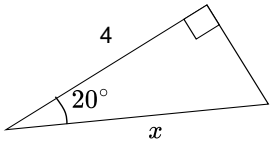
Chapter 9 · Original slide 4

RECAP: Right-Angled Trigonometry

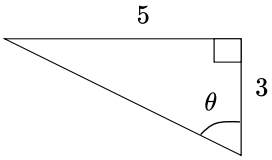
4



You are probably familiar with the formula: $\sin(\theta) = \frac{opp}{hyp}$
But what is the *conceptual* definition of $\sin$?
sin is a function which inputs an angle and gives the ratio between the opposite and hypotenuse.



Find x.



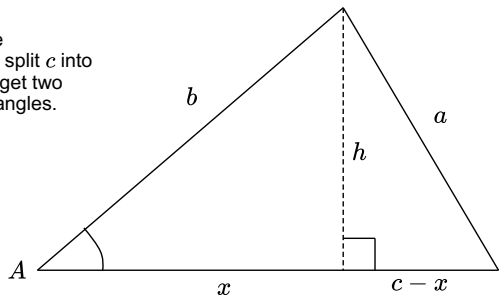
Find θ.

Chapter 9 · Original slide 5

Proof of Cosine Rule

5

We want to use Pythagoras, so split c into two so that we get two right-angled triangles.



By Pythagoras: $h^2 + x^2 = b^2$ and $h^2 + (c-x)^2 = a^2$
Subtracting to eliminate h:
$x^2 - (c-x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$
But $x = b\cos A$ $\therefore$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

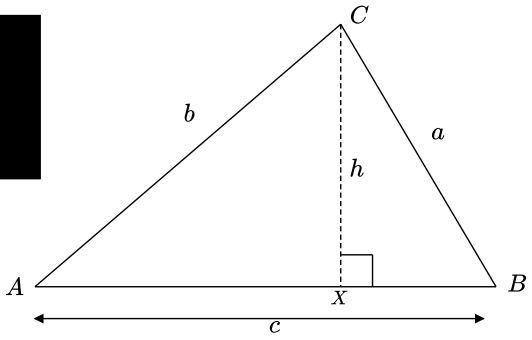
Trigonometry

Chapter 9 · Original slide 6

Proof of Sine Rule

6

The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.



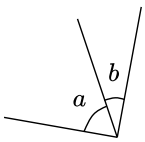
Using basic trigonometry:
$h = a\sin B$ and $h = b\sin A$
$\therefore a\sin B = b\sin A$
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$

Chapter 9 · Original slide 7

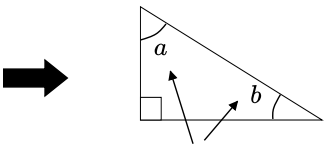
Just for your interest…

7

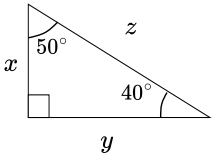
Why does “cosine” contains the word “sine”?



Complementary Angles add to 90°



Therefore these angles are complementary.



Trigonometry

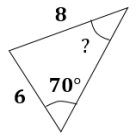
Chapter 9 · Original slide 8

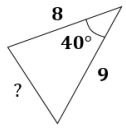
OVERVIEW: Finding missing sides and angles

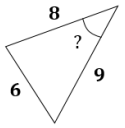
8

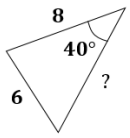
When triangles are not right-angled, we can no longer use simple trigonometric ratios, and must use the cosine and sine rules.

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	
#2 Two sides known and a missing side opposite a known angle	Remaining side	
#3 All three sides	An angle	
#4 Two sides known and a missing side not opposite known angle	Remaining side	







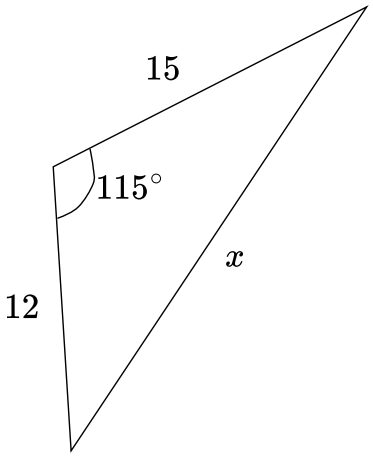


Chapter 9 · Original slide 9

Cosine Rule

9

We use the **cosine rule** whenever we have **three sides** (and an angle) involved.



Proof at end of PowerPoint.

Cosine Rule:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

The only angle in formula is A, so label angle in diagram A, label opposite side a, and so on (b and c can go either way).

Trigonometry

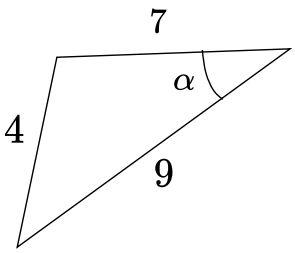
Chapter 9 · Original slide 10

Dealing with Missing Angles

10

You have	You want	Use
All three sides	An angle	Cosine rule

$$\mathbf{a^2 = b^2 + c^2 - 2bc\cos A}$$



Label sides then substitute into formula.

Simplify each bit of formula.

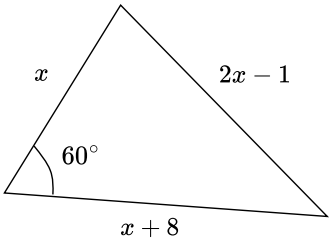
Rearrange "swapping" thing you're subtracting and result

Chapter 9 · Original slide 11

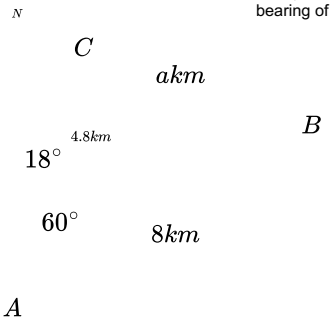
Harder Ones

11

Determine the value of x.



[From textbook] Coastguard station B is 8 km, on a bearing of 060°, from coastguard station A. A ship C is 4.8 km on a bearing of 018°, away from A. Calculate how far C is from B.



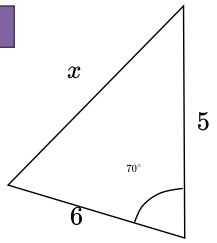
Trigonometry

Chapter 9 · Original slide 12

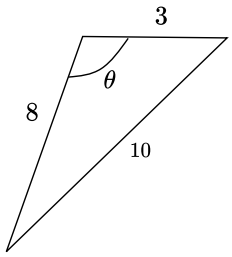
Test Your Understanding

12

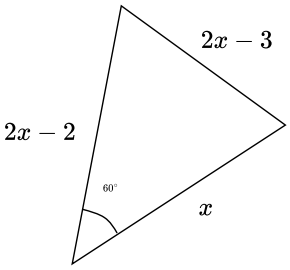
1



2



3

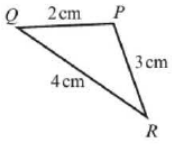


Chapter 9 · Original slide 13

Exercise 9A – Y1 pages 177-179

13

8 Show that $\cos P = -\frac{1}{4}$.



11 The lengths of the sides of a triangle are in the ratio $2 : 3 : 4$.
Calculate the size of the largest angle.

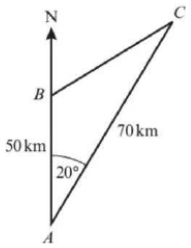
Trigonometry

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Exercise 9A – Y1 pages 177-179

14

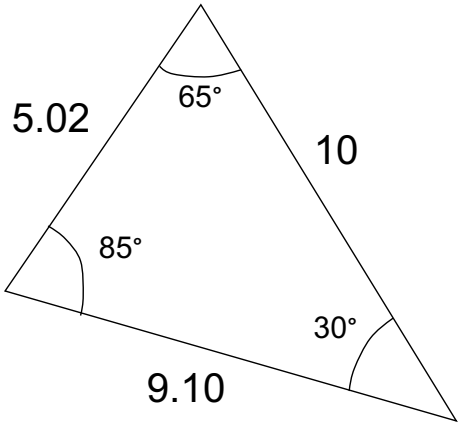
17 The diagram shows three cargo ships, A, B and C, which are in the same horizontal plane. Ship B is 50 km due north of ship A and ship C is 70 km from ship A. The bearing of C from A is $020°$.
a Calculate the distance between ships B and C, in kilometres to 3 s.f. (3 marks)
b Calculate the bearing of ship C from ship B. (4 marks)



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The Sine Rule

16



For this triangle, try calculating each side divided by the sin of its opposite angle. What do you notice in all three cases?

Sine Rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

You have	You want	Use
#1. Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

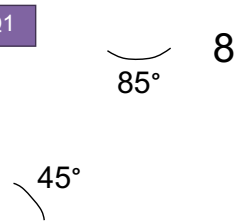
Trigonometry

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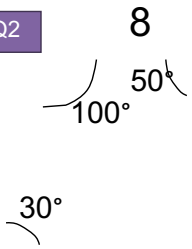
Examples

17

Q1



Q2



You have	You want	Use
4ft. Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

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Examples

18

When you have a missing angle, it's better to reciprocate to get:

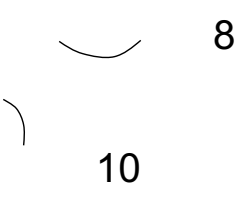
$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

i.e. in general put the missing value in the numerator.

Q3



Q4



Trigonometry

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Exercise 9B – Y1 pages 181-183

19

8 Town B is 6 km, on a bearing of 020°, from town A. Town C is located on a bearing of 055° from town A and on a bearing of 120° from town B. Work out the distance of town C from:
a town A\qquad b town B

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Exercise 9B – Y1 pages 181-183

20

11 In $\triangle ABC$,
$AB = x$ cm, $BC = (4 - x)$ cm, $\angle BAC = y$ and
$\angle BCA = 30^\circ$.
Given that $\sin y = \frac{1}{\sqrt{2}}$, show that

$$x = 4(\sqrt{2} - 1)$$

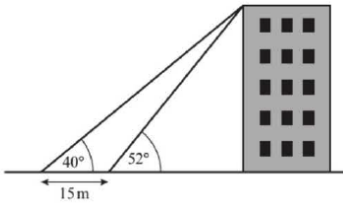
Trigonometry

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Exercise 9B – Y1 pages 181-183

21

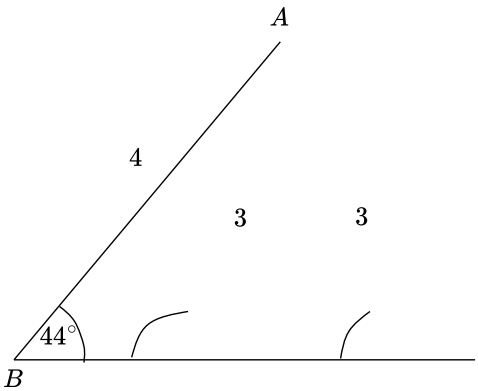
12 A surveyor wants to determine the height of a building. She measures the angle of elevation of the top of the building at two points 15 m apart on the ground.
a Use this information to determine the height of the building. (4 marks)
b State one assumption made by the surveyor in using this mathematical model. (1 mark)



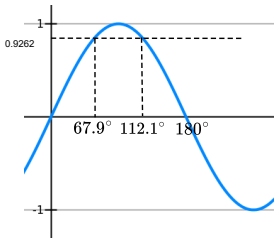
Chapter 9 · Original slide 23

The ‘Ambiguous Case’

23



Suppose you are told that $AB = 4$, $AC = 3$ and $\angle ABC = 44^\circ$. What are the possible values of $\angle ACB$?



Your calculator will give the acute angle of 67.9° (i.e. C_2). But if we look at a graph of sin, we can see there’s actually a second value for $\sin^{-1}(0.9262)$, corresponding to angle C_1.

The sine rule produces two possible solutions for a missing angle:

$$\sin\theta = \sin(180^\circ - \theta)$$

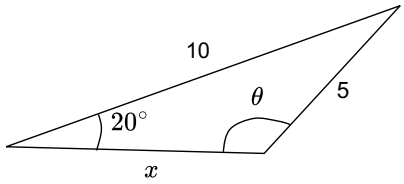
Whether we use the acute or obtuse angle depends on context.

Trigonometry

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Test Your Understanding

24



Given that the angle θ is obtuse, determine θ and hence determine the length of x.

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Exercise 9C – Y1 pages 184-185

25

4 Triangle ABC is such that $AB = 4$ cm, $BC = 6$ cm and $\angle ACB = 36°$. Show that one of the possible values of $\angle ABC$ is $25.8°$ (to 3 s.f.). Using this value, calculate the length of AC.

Trigonometry

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Exercise 9C – Y1 pages 184-185

26

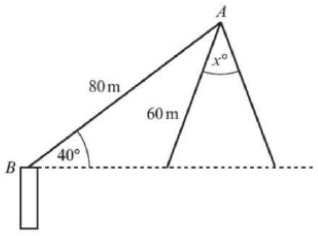
5 Two triangles ABC are such that $AB = 4.5\text{ cm}$, $BC = 6.8\text{ cm}$ and $\angle ACB = 30^\circ$. Work out the value of the largest angle in each of the triangles.

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Exercise 9C – Y1 pages 184-185

27

6 a A crane arm AB of length 80 m is anchored at point B at an angle of 40° to the horizontal. A wrecking ball is suspended on a cable of length 60 m from A. Find the angle x through which the wrecking ball rotates as it passes the two points level with the base of the crane arm at B. (6 marks)
b Write down one modelling assumption you have made. (1 mark)

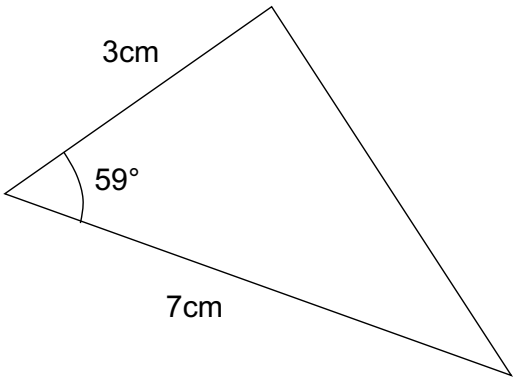


Trigonometry

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Area of Non Right-Angled Triangles

28



$$\text{Area} = \frac{1}{2}ab\sin(C)$$

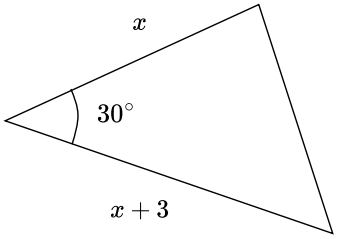
where C is the angle between two sides a and b.

Tip: You shouldn't have to label sides/angles before using the formula. Just remember that the angle is between the two sides.

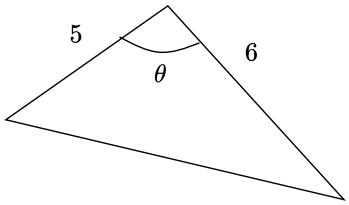
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Test Your Understanding

29



The area of this triangle is 10.
Determine x.



The area of this triangle is also 10.
If θ is obtuse, determine θ.

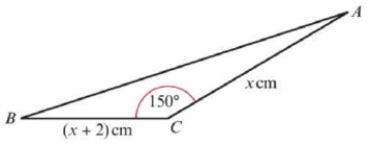
Trigonometry

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Exercise 9D – Y1 pages 186-187

30

4 In triangle ABC, $BC = (x + 2)$ cm, $AC = x$ cm and $\angle BCA = 150^\circ$. Given that the area of the triangle is 5 cm^2, work out the value of x, giving your answer to 3 significant figures.



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Exercise 9D – Y1 pages 186-187

31

5 In $\triangle PQR$, $PQ = (x + 2)$ cm, $PR = (5 - x)$ cm and $\angle QPR = 30^\circ$. The area of the triangle is A cm^2.

a Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

b Use the method of completing the square, or otherwise, to find the maximum value of A, and give the corresponding value of x.

Trigonometry

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Exercise 9D – Y1 pages 186-187

32

6 In $\triangle ABC$, $AB = x$ cm, $AC = (5 + x)$ cm and $\angle BA$
that the area of the triangle is $3\frac{3}{4}$ cm^2
a Show that x satisfies the equation $x^2 + 5x - 15 = 0$.
b Calculate the value of x, giving your answer to 3 significa... figures.

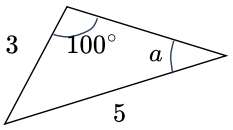
Chapter 9 · Original slide 33

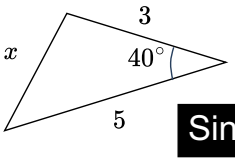
Sin or cosine rule?

33

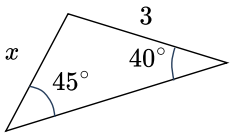
Recall that whenever we have **two "side-angle pairs"** involved, use sine rule. If there's **3 sides** involved, we can use cosine rule. Sine rule is generally easier to use than cosine rule.

Sine Cosine

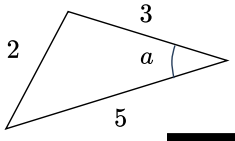




Sine Cosine



Sine Cosine



Sine Cosine

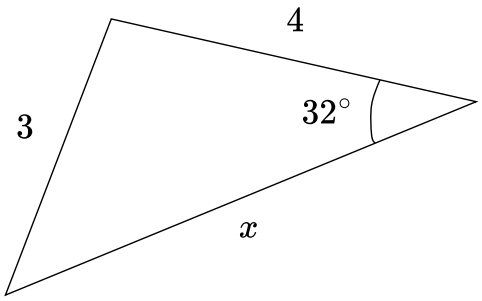
Trigonometry

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Using sine rule twice

34

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice



Given there is just one angle involved, you might attempt to use the cosine rule:

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Using sine rule twice

35

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

1: We could use the sine rule to find this angle.

$$4$$

$$3 \qquad 32^\circ \left($$

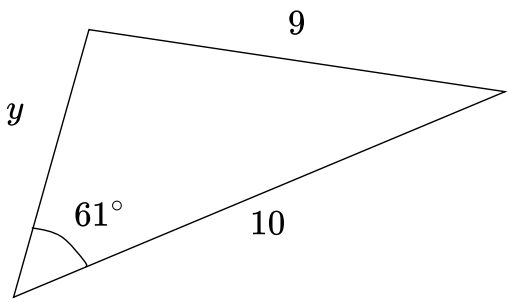
$$x$$

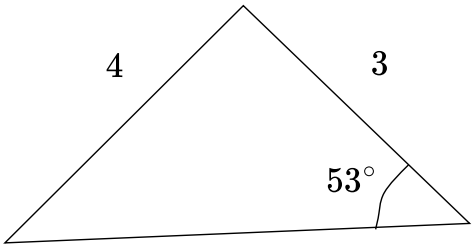
Trigonometry

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Test Your Understanding

36



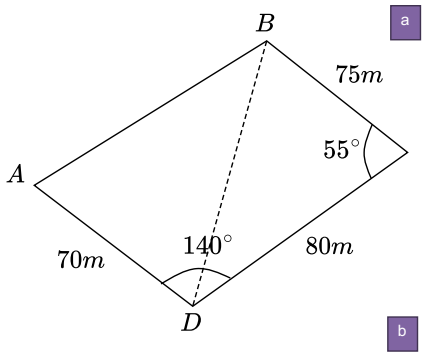


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Problem Solving With Sine/Cosine Rule

37

[From Textbook] The diagram shows the locations of four mobile phone masts in a field, $BC = 75m$. $CD = 80m$, angle $BCD = 55^\circ$ and angle $ADC = 140^\circ$.
In order that the masts do not interfere with each other, they must be at least 70m apart.
Given that A is the minimum distance from D, find:
a) The distance A is from B
b) The angle BAD
c) The area enclosed by the four masts.



a

b

c

Trigonometry

Chapter 9 · Original slide 38

Exercise 9E – Y1 page 189-191

38

12 The perimeter of $\triangle ABC = 15\ \text{cm}$. Given that $AB = 7\ \text{cm}$ and $\angle BAC = 60°$, find the lengths of AC and BC and the area of the triangle.

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Exercise 9E – Y1 page 189-191

39

13 In the triangle ABC, $AB = 14\ \text{cm}$, $BC = 12\ \text{cm}$ and $CA = 15\ \text{cm}$.
a Find the size of angle C, giving your answer to 3 s.f.
b Find the area of triangle ABC, giving your answer in cm^2 to 3 s.f.

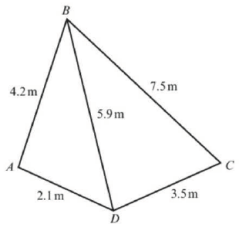
Trigonometry

Chapter 9 · Original slide 40

Exercise 9E – Y1 page 189-191

40

14 A flower bed is in the shape of a quadrilateral as shown in the diagram.
a Find the sizes of angles DAB and BCD. \hfill(4 ma...
b Find the total area of the flower bed. \hfill(3 ma...
c Find the length of the diagonal AC.

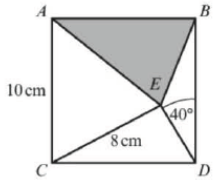


Chapter 9 · Original slide 41

Exercise 9E – Y1 page 189-191

41

15 $ABCD$ is a square. Angle CED is obtuse.
Find the area of the shaded triangle. (7 marks)



Trigonometry

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Sin Graph

44

What does it look like?

Key Features:
• Repeats every $360°$.
• Range: $-1 < \sin(x) < 1$

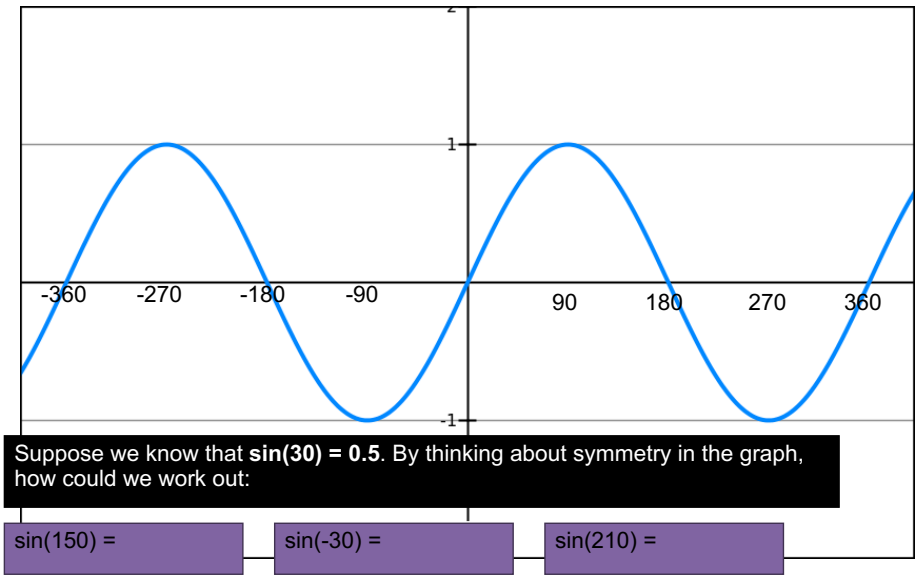
-360 -270 -180 -90 90 180 270 360

Chapter 9 · Original slide 45

Sin Graph

45

What do the following graphs look like?



Suppose we know that **sin(30) = 0.5**. By thinking about symmetry in the graph, how could we work out:

sin(150) = | sin(-30) = | sin(210) =

Trigonometry

Chapter 9 · Original slide 46

Cos Graph

46

What do the following graphs look like?

Key Features:
• Repeats every $360°$.
• Range: $-1 < \sin(x) < 1$

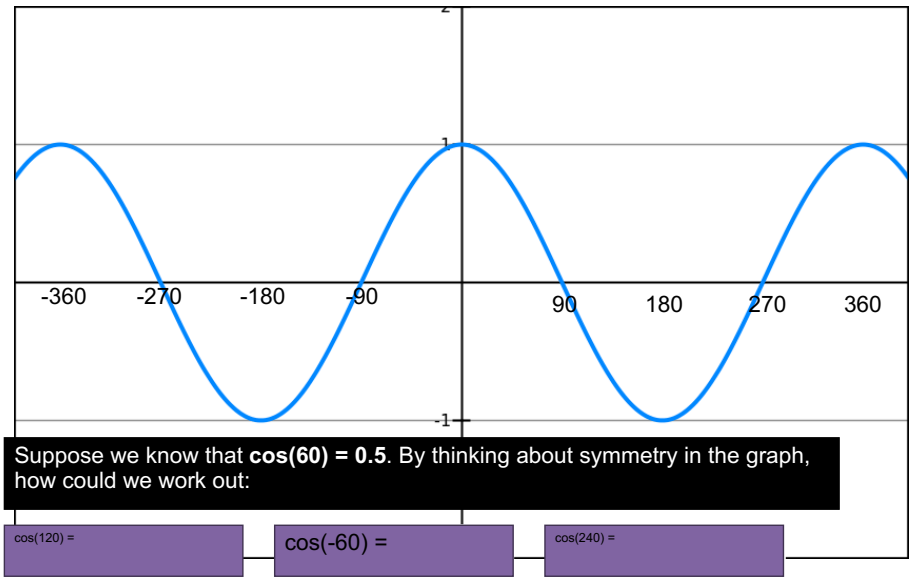
-360 -270 -180 -90 90 180 270 360

Chapter 9 · Original slide 47

Cos Graph

47

What does it look like?



Suppose we know that **cos(60) = 0.5**. By thinking about symmetry in the graph, how could we work out:

cos(120) =

cos(-60) =

cos(240) =

Trigonometry

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Tan Graph

48

What does it look like?



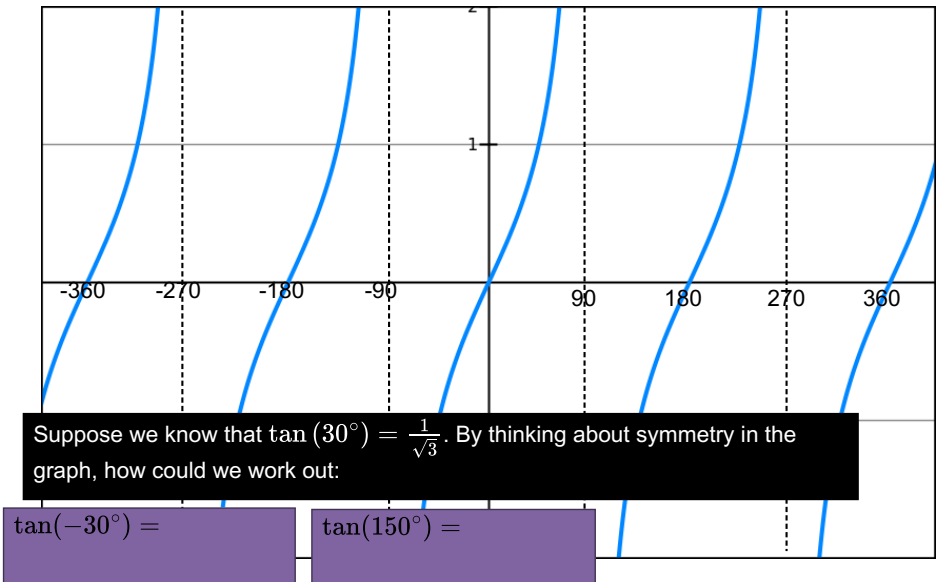
Key Features:
- Repeats every $\mathbf{180}^\circ$.
- Roots: $0, 180^\circ, -180^\circ, \ldots$
- Range: $\tan(x) \in \mathbb{R}$ (i.e. no min/max value!)
- Asymptotes: $x = \pm 90^\circ, \pm 270^\circ, \ldots$

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Tan Graph

49

What does it look like?



Suppose we know that $\tan(30^\circ) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

$\tan(-30^\circ) =$

$\tan(150^\circ) =$

Trigonometry

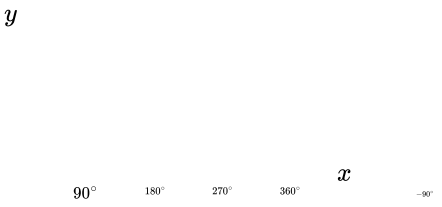
Chapter 9 · Original slide 50

Transforming Trigonometric Graphs

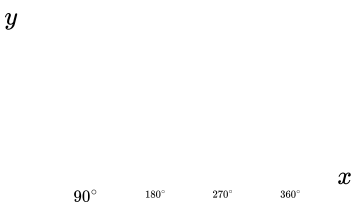
50

There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs ‘inside’ the function (i.e. input modified) or ‘outside’ the function (i.e. output modified).

Sketch $y = 4\sin x,\ 0 \leq x \leq 360^\circ$



Sketch $y = \cos(x + 45^\circ),\ 0 \leq x \leq 360^\circ$

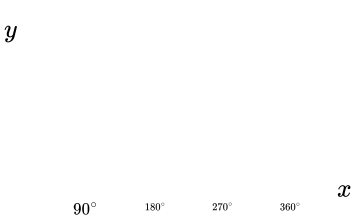


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Transforming Trigonometric Graphs

51

Sketch $y = -\tan x,\ 0 \leq x \leq 360^\circ$



Sketch $y = \sin\left(\frac{x}{2}\right),\ 0 \leq x \leq 360^\circ$



Trigonometry

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Exercise 9F – Y1 Page 194

52

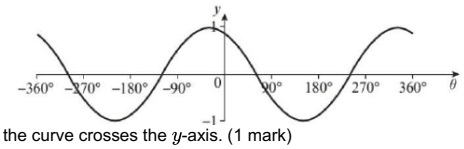
1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

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Exercise 9G – Y1 Pages 197-198

53

7 The graph shows the curve
$y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)



b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

b $\cos 30^\circ = \frac{\sqrt{3}}{2}$; $\left(0, \frac{\sqrt{3}}{2}\right)$

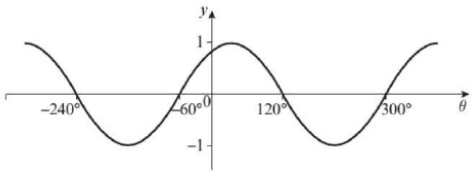
Trigonometry

Chapter 9 · Original slide 54

Exercise 9G – Y1 Pages 197-198

54

8 The graph shows the curve with equation $y = \sin(x + k), -360^\circ \leq x \leq 360^\circ$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)



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Exercise 9G – Y1 Pages 197-198

55

9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?

Trigonometry

Chapter 10 · Original slide 3

sin/cos/tan of $30^\circ, 45^\circ, 60^\circ$

3

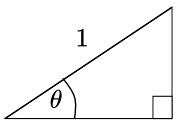
You will frequently encounter angles of 30°, 60°, 45° in geometric problems. Why?

Chapter 10 · Original slide 4

The Unit Circle and Trigonometry

4

For values of θ in the range $0 < \theta < 90^\circ$, you know that $\sin\theta$ and $\cos\theta$ are lengths on a right-angled triangle:



Trigonometry

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Mini-Exercise

5

Use the unit circle to determine each value in the table, **using either "0", "+ve", "-ve", "1", "-1" or "undefined"**.
Recall that the point on the unit circle has coordinate $(\cos\theta, \sin\theta)$ and OP has gradient $\tan\theta$.

	$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 0$ 	1	0	0
$0 < \theta < 90^\circ$ 			
$\theta = 90^\circ$ 			
$90^\circ < \theta < 180^\circ$ 			

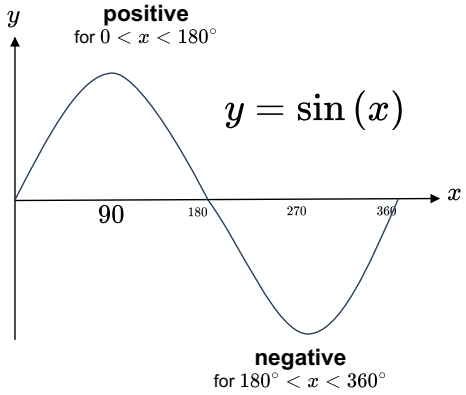
	$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 180^\circ$ 			
$180^\circ < \theta < 270^\circ$ 			
$\theta = 270^\circ$ 			
$270^\circ < \theta < 360^\circ$ 			

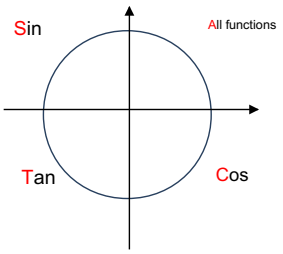
Chapter 10 · Original slide 6

The Unit Circle and Trigonometry

6

The unit circles explains the behaviour of these trigonometric graphs beyond 90°.
However, the easiest way to remember whether $\sin(x), \cos(x), \tan(x)$ are positive or negative is to just do a **very quick sketch (preferably mentally!)** of the corresponding graph.





Note: The textbook uses something called '*CAST diagrams*', which is an acronym for which quadrants the **C**osine, **A**ll, **S**ine and **T**angent are positive

Trigonometry

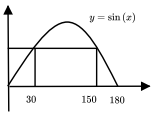
Chapter 10 · Original slide 7

A Few Trigonometric Angle Laws

7

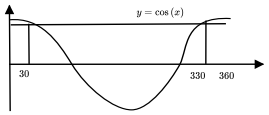
The following are all easily derivable using a quick sketch of a trigonometric graph, and are merely a convenience so you don't always have to draw out a graph every time.
You are highly encouraged to **memorise these** so that you can do exam questions faster.

1. $\sin(x) = \sin(180^\circ - x)$
   e.g. $\sin(150^\circ) = \sin(30^\circ)$

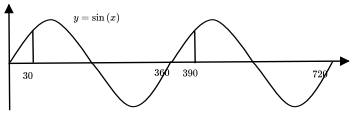


We saw this in the previous chapter when covering the 'ambiguous case' when using the sine rule.

2. $\cos(x) = \cos(360^\circ - x)$
   e.g. $\cos(330^\circ) = \cos(30^\circ)$



3. sin and cos repeat every 360° but tan every 180°
   e.g. $\sin(390^\circ) = \sin(30^\circ)$



4. $\sin(x) = \cos(90^\circ - x)$
   e.g. $\sin(50^\circ) = \cos(40^\circ)$

Remember from the previous chapter that "cosine" by definition is the sine of the "complementary" angle. This was/is never covered in the textbook but caught everyone by surprise when it came up in a C3 exam.

Chapter 10 · Original slide 8

Examples

8

Without a calculator, work out the value of each below.

$\tan(225^\circ) =$

$\tan(210^\circ) =$

$\sin(150^\circ) =$

$\cos(300^\circ) =$

$\sin(-45^\circ) =$

$\cos(750^\circ) =$

$\cos(120^\circ) =$

Use the 'laws' where you can, but otherwise just draw out a quick sketch of the graph.

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every 360° but tan every 180°

Trigonometry

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Test Your Understanding

9

Without a calculator, work out the value of each below.

$$\cos(315^\circ) =$$

$$\sin(420^\circ) =$$

$$\tan(-120^\circ) =$$

$$\tan(-45^\circ) =$$

$$\sin(135^\circ) =$$

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every 360° but tan every 180°

Chapter 10 · Original slide 10

Exercise 10B – Y1 Page 209

10

1 Express the following as trigonometric ratios of either 30°, 45° or 60°, and hence find their exact values.\quad
a $\sin 135^\circ$\quad**b** $\sin(-60^\circ)$\quad**c** $\sin 330^\circ$\quad**d** $\sin 420^\circ$\quad**e** $\sin(-300^\circ)$\quad**f** $\cos 120^\circ$\quad**g** $\cos 300^\circ$\quad**h** $\cos 225^\circ$\quad**i** $\cos(-210^\circ)$\quad**j** $\cos 495^\circ$\quad**k** $\tan 135^\circ$\quad**l** $\tan(-225^\circ)$\quad**m** $\tan 210^\circ$\quad**n** $\tan 300^\circ$\quad**o** $\tan(-120^\circ)$.

t

t

Trigonometry

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Exercise 10A/B – Y1 Page 207, 209

11

Challenge

The diagram shows an isosceles right-angled triangle ABC.

$AE = DE = 1$ unit. Angle $ACD = 30°$.

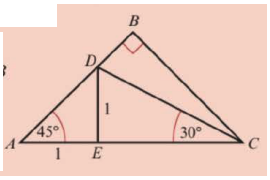
a Calculate the exact lengths of

i CE \quad **ii** DC \quad **iii** BC \quad **iv** DB

b State the size of angle BCD.

c Hence find exact values for

i $\sin 15°$ \quad **ii** $\cos 15°$

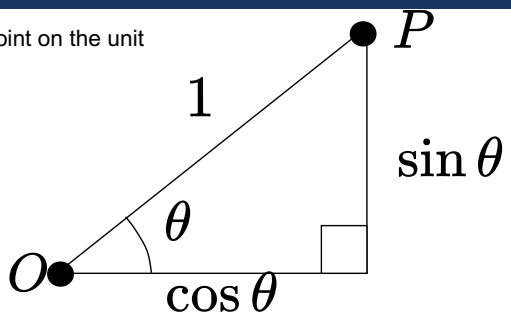


Chapter 10 · Original slide 12

Trigonometric Identities

12

Returning to our point on the unit circle…



1

2

Pythagoras gives you...

$\sin^2\theta$ is a shorthand for $(\sin\theta)^2$. It does NOT mean the sin is being squared – this does not make sense as sin is a function, and not a quantity that we can square!

Trigonometry

Chapter 10 · Original slide 13

Application of identities #1: Proofs

13

Prove that $1 - \tan\theta \sin\theta \cos\theta \equiv \cos^2\theta$

Tip #1: Turn any tan's into sin's and cos's.

$$\tan x = \frac{\sin(x)}{\cos(x)}$$
$$\sin^2 x + \cos^2 x = 1$$

Recall that $\equiv$ means 'equivalent to', and just means the LHS is **always** equal to the RHS for all values of θ.
From Chapter 7 ('Proofs') we saw that usually the best method is to manipulate one side (e.g. LHS) until we get to the other (RHS).

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More Examples

14

Edexcel C2 June 2012 Paper 1 Q16

Prove that $\tan\theta + \frac{1}{\tan\theta} \equiv \frac{1}{\sin\theta\cos\theta}$

Simplify $5 - 5\sin^2\theta$

Tip #2: In any addition/subtraction involving at least one fraction (with trig functions), always combine algebraically into one.

Trigonometry

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Test Your Understanding

15

Prove that $\frac{\tan x \cos x}{\sqrt{1-\cos^2 x}} \equiv 1$

Prove that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

AQA IGCSE Further Maths Worksheet

Prove that $\tan^2\theta \equiv \frac{1}{\cos^2\theta} - 1$

Chapter 10 · Original slide 16

Exercise 10C – Y1 Page 211-212

16

10 In each of the following, eliminate θ to give an equation relating x and y:

a $x = \sin\theta, y = \cos\theta$ **b** $x = \sin\theta, y = 2\cos\theta$

c $x = \sin\theta, y = \cos^2\theta$ **d** $x = \sin\theta, y = \tan\theta$

Trigonometry

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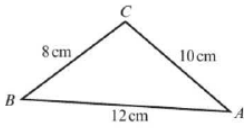
Exercise 10C – Y1 Page 211-212

17

11 The diagram shows the triangle ABC with $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.

a Show that $\cos B = \frac{9}{16}$. \hfill (3 ma

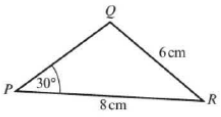
b Hence find the exact value of $\sin B$.



12 The diagram shows triangle PQR with $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and angle $QPR = 30°$.

a Show that $\sin Q = \frac{2}{3}$.

b Given that Q is obtuse, find the exact value of $\cos Q$.



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Solving Trigonometric Equations

19

Remember those trigonometric angle laws (on the right) earlier this chapter? They're about to become very useful!

Reminder of 'trig laws':
- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every $360°$ but tan every $180°$

Solve $\sin\theta = \frac{1}{2}$ in the interval $0 \leq \theta \leq 360°$.

Solve $5\tan\theta = 10$ in the interval $-180° \leq \theta < 180°$

Trigonometry

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Slightly Harder Ones…

20

Solve $\sin\theta = -\frac{1}{2}$ in the interval $0 \leq \theta \leq 360^\circ$.

In general you should have 2 solutions per 360° (except when at a peak or trough of the trig graph)

Note that we've had to use a second law, i.e. that $\sin$ repeats every 360°.

Solve $\sin\theta = \sqrt{3}\cos\theta$ in the interval $0 \leq \theta \leq 360^\circ$.

Hint: The problem here is that we have two different trig functions. Is there anything we can divide both sides by so we only have one trig function?

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Test Your Understanding

21

Solve $2\cos\theta = \sqrt{3}$ in the interval $0 \leq \theta \leq 360^\circ$.

Solve $\sqrt{3}\sin\theta = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.

Trigonometry

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Exercise 10D – Y1 215-216

22

7 A teacher asks two students to solve the equation $2\cos x = 3\sin x$ for $-180^\circ \leq x \leq 180^\circ$. The attempts are shown:
Student A: $\tan x = \frac{3}{2}$, $x = 56.3^\circ$ or $x = -123.7^\circ$.
Student B: $4\cos^2 x = 9\sin^2 x$, $4(1 - \sin^2 x) = 9\sin^2 x$, $4 = 13\sin^2 x$, $\sin x = \pm\sqrt{\frac{4}{13}}$, $x = \pm 33.7^\circ$ or $x = \pm 146.3^\circ$.
a Identify the mistake made by Student A. (1 mark)
b Identify the mistake made by Student B and explain the effect it has on their solution. (2 marks)
c Write down the correct answers to the question. (1 mark)

9 Find all the values of θ, to 1 decimal place, in the interval $0 < \theta < 360^\circ$ for which $\tan^2\theta = 9$

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Exercise 10D – Y1 215-216

23

10 a Show that $4\sin^2 x - 3\cos^2 x = 2$ can be written as $7\sin^2 x = 5$.
b Hence solve, for $0 \leq x \leq 360^\circ$, the equation $4\sin^2 x - 3\cos^2 x = 2$. Give your answers to 1 decimal place.

11 a Show that the equation $2\sin^2 x + 5\cos^2 x = 1$ can be written as $3\sin^2 x = 4$.
b Use your result in part a to explain why the equation $2\sin^2 x + 5\cos^2 x = 1$ has no solutions.

Trigonometry

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Harder Equations

24

Harder questions replace the angle θ with a linear expression.

Solve $\cos 3x = -\frac{1}{2}$ in the interval $0 \leq x \leq 360^\circ$.

STEP 1: Adjust the range of values for θ to match the expression inside the cos.

STEP 2: Immediately after applying an inverse trig function (and BEFORE dividing by 3!), find all solutions up to the end of the interval.

STEP 3: Then do final manipulation to each value.

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Further Examples

25

Solve $\sin(2x + 30^\circ) = \frac{1}{\sqrt{2}}$ in the interval $0 \leq x \leq 360^\circ$.

To get from x to $2x + 30^\circ$ we double and add 30°. So do the same to the upper and lower bound!

Solve $\sin x = 2\cos x$ in the interval $0 \leq x < 300^\circ$

By dividing both sides by $\cos x$, the $\sin x$ becomes $\tan x$ and the $\cos x$ disappears, leaving a trig equation helpfully only in terms of one trig function.

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Test Your Understanding

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Solve, for $0 \leq x < 180^\circ$,
$\cos(3x - 10^\circ) = -0.4$,
giving your answers to 1 decimal place. You should show each step in your working. (7)

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Exercise 10E – Y1 p219

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4 Solve for $0 \leq x \leq 180^\circ$ the equations:
a $\sin(x + 20^\circ) = \frac{1}{2}$.
b $\cos 2x = -0.8$, giving your answers to 1 decimal place.

5 a Sketch for $0 \leq x \leq 360^\circ$ the graph of $y = \sin(x + 60^\circ)$.
b Write down the exact coordinates of the points where the graph meets the coordinate axes.
c Solve, for $0 \leq x \leq 360^\circ$, the equation $\sin(x + 60^\circ) = 0.55$, giving your answers to 1 decimal place.

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Exercise 10E – Y1 p219

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6 a Given that $4\sin x = 3\cos x$, write down the value of $\tan x$.
b Solve, for $0 \leq \theta \leq 360^\circ$, $4\sin 2\theta = 3\cos 2\theta$ giving your answers to 1 decimal place.

7 The equation $\tan kx = -\frac{1}{\sqrt{3}}$, where k is a constant and $k > 0$, has a solution at $x = 60^\circ$.
a Find a possible value of k.
b State, with justification, whether this is the only such possible value of k.

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Quadratics in sin/cos/tan

29

We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360^\circ$.

Method 1: Use a substitution.

Method 2: Factorise without substitution.

This is the same, but we 'imagine' $\sin x$ as a single variable and hence factorise immediately.

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More Examples

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Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360°$.

Missing the negative case would result in the loss of multiple marks. Beware!

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180° \leq x \leq 180°$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

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Test Your Understanding

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(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360°$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

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Exercise 10F – Y1 P221-222

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5 Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

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Exercise 10F – Y1 P221-222

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7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

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Exercise 10F – Y1 P221-222

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9\quad $\tan^2 x - 2\tan x - 4 = 0$.
a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.
b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540^\circ$.